**Ref:** CTU/N/00/CMETS_NR/28

**Date:** 08-04-2024

**As per distribution list**

**Subject: 28th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting**

Dear Sir/Ma'am,

Please find enclosed the minutes of the 28th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 27th March, 2024 (Wednesday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

08/04/24

**(Kashish Bhambhani)**
**General Manager (CTU)**

**Distribution List:**

| | |
|---|---|
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| **Director (Operations)**<br>Delhi Transco Ltd.<br>Shakti Sadan, Kotla Road,<br>New Delhi-110 002 | **Director (Technical)**<br>Haryana Vidyut Prasaran Nigam Ltd.<br>Shakti Bhawan, Sector-6,<br>Panchkula-134109, Haryana |
| **Executive Director (AM)**<br>POWERGRID<br>Saudamini, Plot No.2, Power Grid Corporation Of India Limited, Near, August Kranti Marg, Sector 29,<br>Gurugram, Haryana 122001 | **Executive Director (Engg.)**<br>POWERGRID<br>Saudamini, Plot No.2, Power Grid Corporation Of India Limited, Near, August Kranti Marg, Sector 29,<br>Gurugram, Haryana 122001 |

## Minutes of 28th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 27.03.2024

The 28th Consultation Meeting for Evolving Transmission Schemes in the Northern Region (CMETS-NR) was held through VC on 27.03.24. List of participants is enclosed at **Annexure-A**.

### A. Network Expansion Scheme

### A1. Transmission system for evacuation of power from Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-4 :3.5GW)

It was stated that Transmission system for evacuation of power from Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-4 :3.5GW) was agreed in 27th CMETS-NR meeting held on 10.01.24. Subsequently, Line lengths of some of the transmission lines were modified w.r.t new location of Barmer-I PS as well as review in Gati Shakti portal. Accordingly in view of high voltages in off solar hours, reactive compensation of some of the lines was also proposed to be modified. Details in this regard are as under :

| S. No. | Agreed Scope as per 27th CMETS-NR | Modified Scope |
|---|---|---|
| 1 | Barmer-I PS – Merta-II 765 kV D/c line along with 240 MVAr switchable line reactor at Barmer-I PS end and 330MVAr switchable line reactor at Merta-II PS end for each circuit of Barmer-I PS – Merta-II 765 kV D/c line (~320 km) | Barmer-I PS – Merta-II 765 kV D/c line along with 330 MVAr switchable line reactor for each circuit at each end of Barmer-I PS – Merta-II 765 kV D/c line (~345 km)<br><br>110MVAr spare reactor unit at Barmer-I PS (single phase) |
| 2 | Dausa - Ghiror 765 kV D/c line along with 240 MVAr switchable line reactor for each circuit at each end of Dausa - Ghiror 765 kV D/c line (~285 km) | Dausa - Ghiror 765 kV D/c line along with 330MVAr switchable line reactor at Ghiror end and 240 MVAr switchable line reactor at Dausa end for each circuit of Dausa - Ghiror 765 kV D/c line (~305 km)<br><br>110MVAr spare reactor unit at Ghiror S/s (single phase) |

Grid-India & CEA agreed to the above proposal. Further, LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror along with 240 MVAr switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line may be considered as LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) **2xS/c** line (in view of 2xS/c line configuration) at Ghiror along with 240 MVAr switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line. Grid-India stated that with LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror S/s, utilization of 240MVAr line reactor at Agra end on 765kV Agra-Ghiror section may be reviewed. CTU stated that with proposed LILO at Ghiror S/s, 765kV Agra -Ghiror section will be about 130kms and it view of that it is recommended that line be reactor may be kept in above section. Grid-India agreed for the same

No other observations were received on above proposals. In view of that, above modifications were agreed.

## A2. Non-compliance of N-1 Contingency at 400/220kV Bassi & 400/220 kV Bhiwadi S/s

It was stated that in 213th OCC meeting held on 22.11.23, agenda of non-compliance of transformer N-1 loadings in some critical Grid Substations i.e.400/220kV Bassi, 400/220kV Bhiwadi, 400/220kV Jaipur (South), 400/220kV Neemrana S/s & 765/400kV Bhiwani S/s of Northern region was deliberated. In the meeting NRLDC presented the loading pattern of ICT's of these aforementioned Grid Stations of Northern region. Subsequently, it was inferred that loading at ICT's at Bassi S/s is on higher side resulting in non-compliant of N-1 Contingency while for Bhiwadi S/s also loading pattern is marginally higher. Whereas in case of Jaipur South S/s and Neemrana S/s loading is within limits. Rajasthan SLDC representative mentioned that they have planned LILO of 220kV Dausa-Sawai Madhopur line at Jaipur South S/s and thereafter there will be requirement of additional ICT only at Jaipur (South) S/s not at Bassi S/s.

NRPC asked POWERGRID to formally submit their request to CTU for review of N-1 Contingency loading criteria in aforementioned critical Grid Substations of Northern Region and subsequently matter may be taken up in the CMETS meeting of CTU. Subsequently, POWERGRID vide letter dated 21.12.23 reiterated to review the requirement of additional ICTs at Jaipur South, Bassi, Neemrana, Bhiwadi & Bhiwani substation of Northern region as per deliberation in 213th OCC meeting.

At present, based on deliberation in above OCC meeting, requirement for ICT augmentation at 400/220kV Bassi & 400/220kV Bhiwadi is analyzed. Details are as under

### 1. Augmentation of 500 MVA (4th) ICT at 400/220 kV Bassi substation

It was stated that at present, 400/220kV Bassi Substation has 2x315+1x500MVA ICTs thus makes total transformation capacity of 1130MVA. POWERGRID in 203rd OCC stated that the loading at ICTs at Bassi is more than 900MW resulting in non-compliance of N-1 Contingency.

From the loading pattern of Bassi ICTs (2x315+1x500) of past one year, it is observed that cumulative maximum loading on 400/220 kV ICTs is about 1100 MW and loading is higher (>850MW) in winter season (Oct'23-Jan'24) for sufficient duration of time. From the analysis, it emerged that loading is breaching N-1 limit for peak Loading of ICTs (total 855 MW on 3 nos. ICTs). Further in CTU planning file for 2026-27 timeframe, ICT loadings are higher (312MW in N-1 Contingency) and may breach the N-1 limit.

As per deliberation held in 213th OCC meeting, with planned a LILO of 220kV Dausa-Sawai Madhopur line at Jaipur South S/s, loading of Bassi ICT is relived marginally (30MW/ICT in N-1 contingency), however 400/220 kV ICTs will still be non-compliant in future. The loading pattern of Bassi ICTs for past one year is as below

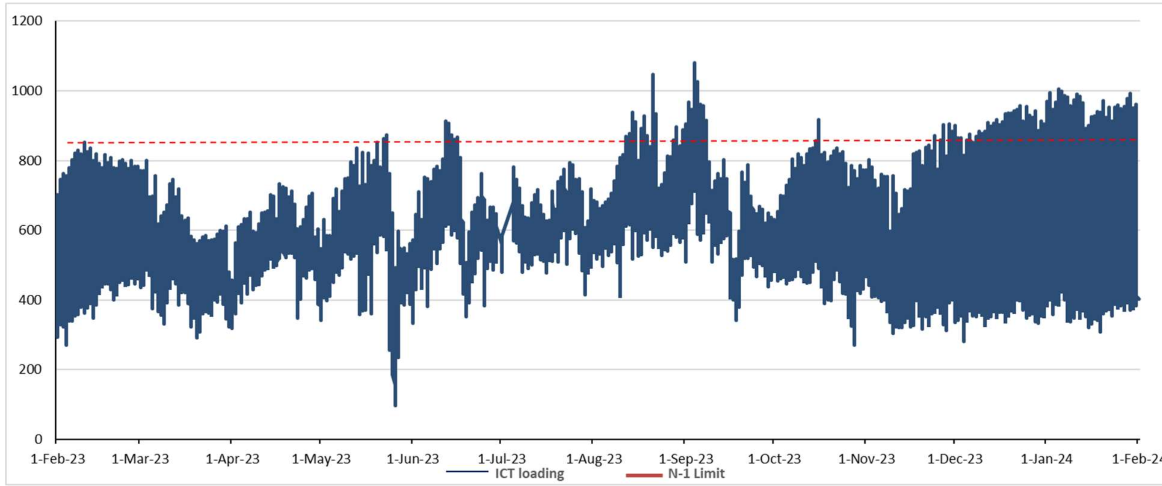


**Fig 1: Loading of Bassi ICTs of Past one year (Source: Grid-India)**

POWERGRID vide mail dated 22.02.24 confirmed space availability for Augmentation of 400/220kV transformer ($4^{th}$,1x500MVA) along with transformer bays at 400/220kV Bassi (PG) S/s.

In the meeting, Grid-India stated that at Bassi there is urgent requirement of augmentation of 400/220kV ICT as ICTs were N-1 non-compliant in current winter season. RVPN stated that at present total 6 nos. 220kV feeders are connected to Bassi substation with peak drawl requirement of 1200-1300MW and as per loading pattern, loading is already breaching N-1 limit. Further, with Hindaun ICT and LILO of 220kV Dausa-Sawai Madhopur line at Jaipur South, ICT loading at Bassi may reduce to only some extent. RVPN further stated that considering future drawl requirement at Bassi S/s, ICT augmentation at Bassi may be considered.

In view of above deliberations, following ICT augmentation scheme was agreed in ISTS:

- Augmentation of 400/220 kV, 500 MVA ($4^{th}$) ICT at Bassi (PG) S/s along with associated transformer bays

## 2. <u>Augmentation of 500 MVA (4th) ICT at 400/220 kV Bhiwadi (Hybrid) substation</u>

At present, 400/220kV Bhiwadi Substation has 3X315 MVA ICTs (total transformation capacity of 945MVA). POWERGRID in 203rd OCC stated that the loading at ICTs at Bhiwadi is more than 800MW resulting in non-compliance of N-1 Contingency.

From the loading pattern of Bhiwadi ICTs (3x315 MVA) , it is observed that cumulative maximum loading on 400/220 kV ICTs is about 820 MW and loading is higher (>600MW) in Jun-Sep'23 & Jan'24-Feb'24 for sufficient duration of time. From the analysis, it emerged that loading is breaching N-1 limit for peak Loading of ICTs (800 MW). Further in CTU planning file, ICT loadings are higher (310 MW in N-1 Contingency) and may breach the N-1 limit The loading pattern of Bhiwadi ICTs for past one year is as below (Fig 2):

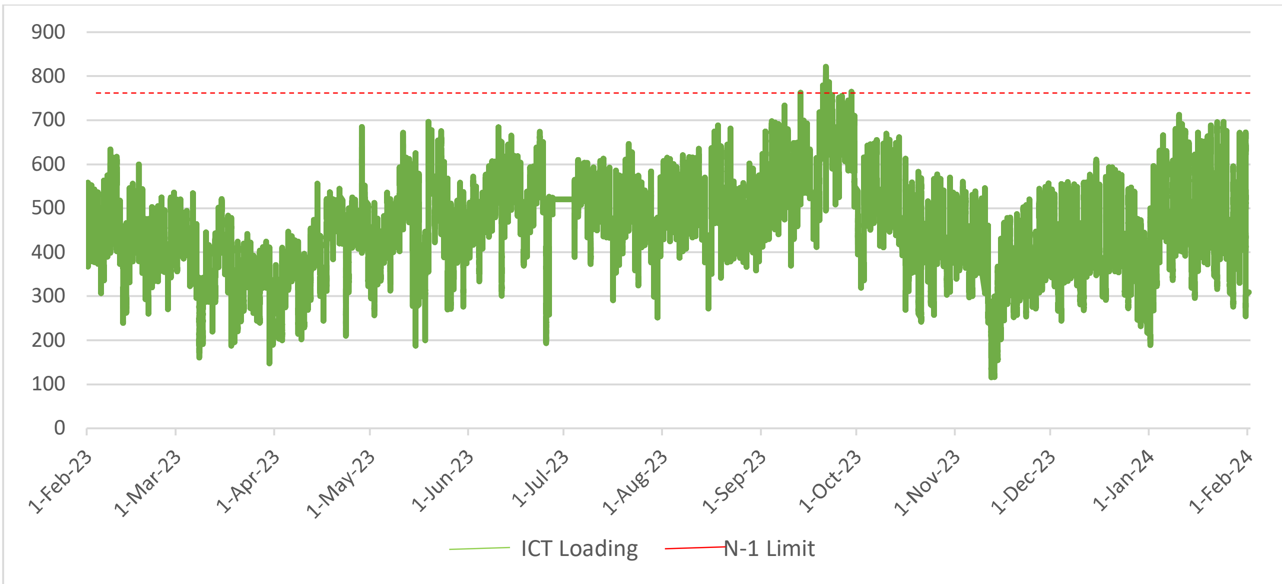


**Fig 2: Loading of Bhiwadi ICTs of Past one year (Source: Grid-India)**

POWERGRID vide mail dated 23.02.24 confirmed space availability for Augmentation of 400/220kV transformer (4th,1x500MVA) along with transformer bays at 400/220kV Bhiwadi (PG) S/s. However, 220kV. Side of ICT is required to be routed through cable.

In the meeting Grid-India stated that at Bhiwadi transformer loading are close to N-1 limit and in view of that requirements may be assessed based on future load growth in Bhiwadi complex considering planned interconnections by RVPN/HVPN in future. RVPN stated that at present for

Rajasthan, 4 nos. of 220kV feeders are interconnected at Bhiwadi substation and no new interconnection is planned for drawl in future. RVPN also stated that a new 400/220kV substation in Bhiwadi is recently approved by RVPN board to cater future drawl requirement at Bhiwadi complex in view of anticipated load growth. In reply to implementation timeframe of proposed new substation in Bhiwadi, RVPN stated that implementation may take minimum 3 years from now.

POWERGRID stated that loading at Bhiwadi substation are critical and N-1 noncompliance in peak load condition. Further, 2 nos. of 400/220kV ICTs (make 2003 year) already completed 21 years of life and Gases are high due to high loading of ICTs. Considering above ICT augmentation is required at Bhiwadi substation. CTU stated that ICT loadings are close to N-1 limit, however HVPN/RVPN need to concur on requirements of ICT augmentation. CTU stated that decision of ICT augmentation may be expedited as Implementation of 400/220kV ICT may take 18-21 months from award and ICT may be available by mid of 2026. Representative from HVPN were not present in the meeting.

RVPN stated that they will re-assess the load pattern at 400/220kV Bhiwadi substation along with future load growth requirement in complex and therefore requested to take up agenda in next meeting along with HVPN views for their drawl requirement.

Considering above, it was decided that agenda for augmentation at 400/220kV Bhiwadi S/s will be taken up in next CMETS-NR meeting.

**A3. <u>Reactive Compensation of Transmission scheme for evacuation of power from Ratle HEP (850MW)</u>**

It was stated that Transmission scheme for Ratle HEP (850MW) was agreed in 26th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 01.12.2023. The scope of earlier agreed scheme is as under

**<u>Common Transmission system for Connectivity under GNA (under ISTS) :</u>**

1. 400 kV Kishenpur-Samba D/c line (Quad) (2nd) - 35kms
2. Bypassing of 400kV Kishtwar – Kishenpur line (Quad) at Kishenpur and connecting it with one of the circuit of Kishenpur-Samba 400kV D/c line(Quad), thus forming 400kV Kishtwar-Samba (Quad) direct line (one ckt)
3. Termination of another circuit of 400 kV Kishenpur-Samba D/c line(Quad) at Kishenpur at 400kV bay vacated at Kishenpur (After bypassing of one ckt of 400 kV Kishtwar-Kishenpur at Kishenpur) –165km
4. Bypassing both ckts of 400kV Kishenpur – Samba D/c line (Twin) & 400 kV Samba – Jalandhar D/c line (Twin) at Samba and connecting them together to form Kishenpur– Jalandhar D/c direct line (Twin)
5. 400kV Samba- Jalandhar D/c line(Quad) (only one circuit is to be terminated at Jalandhar while second circuit would be connected to bypassed circuit of Jalandhar –Nakodar 400kV D/c line) – 170km
6. Bypassing 400kV Jalandhar – Nakodar line (Quad) at Jalandhar and connecting it with one of the circuit of Samba-Jalandhar 400kV D/c line(Quad Moose), thus forming Samba –Nakodar line
7. Termination of another circuit of 400 kV Samba -Jalandhar D/c line (Quad) at Jalandhar at bay vacated at Jalandhar (After bypassing of one ckt of 400 kV Jalandhar - Nakodar at Jalandhar)-177km

8. LILO of 400 kV Kishenpur- Dulhasti line (Twin Zebra) at Kishtwar S/s
9. Reconductoring of 400 kV Kishenpur-Kishtwar line* (Twin HTLS) (minimum 2100 MVA capacity) (formed after LILO of Kishenpur- Dulhasti line at Kishtwar S/s) along with bay upgradation works (2000A to 3150A) at Kishenpur end for above line

**** Conductor configuration for reconductoring of 400 kV Kishenpur-Kishtwar line will be finalized based on POWERGRID inputs.***

***Line reactors will be finalized based on POWERGRID/Sterlite inputs***

Subsequently, CTU vide mail 20.11.23 to POWERGRID requested information regarding maximum rating of the HTLS conductor suitable for Reconductoring of 400 kV Kishenpur-Kishtwar line (Twin Zebra), considering altitude and the bundle size of the line. CTU vide mail 26.12.23 to POWERGRID informed that the minimum Ampacity requirement is 1516A/ckt (2100MVA/ckt) for Reconductoring of 400 kV Kishenpur-Kishtwar line (Twin Zebra) (formed after LILO of Kishenpur- Dulhasti line at Kishtwar S/s) and requested to provide the conductor configuration for reconductoring of above line in order to avoid Corona inception gradient at high altitude. POWERGRID vide mail 26.02.24 informed that Conductor configuration shall remain the same as that of existing transmission line i.e. twin bundle of conductor diameter-28.62mm, However, information of highest and average altitude is being taken up with, details for the same will be forwarded separately.

Further in the meeting, Grid-India stated that the requirement of line reactors for lines such as 400kV Kishtwar-Kishenpur (132km) and 400kV Kishtwar-Samba (160km) may be studied given the persistent issues of high voltage in hydro complex in winter months during off-peak hours. CTU informed that requirement of line reactor will be reviewed considering space availability. Accordingly, CTU analysed the reactive compensation requirement, however feasibility for installation of line reactors along with space confirmation from POWERGRID & Sterlite iwassought.

Subsequently, POWERGRID provided the feasibility and space availability at Kishnepur, Samba & Jallandhar S/s. POWERGRID informed that for installation of line reactor at Samba end of 400kV Kishtwar-Samba 400kV line(Quad), no space is available. This is primarily due to the presence of a dead-end tower of 400kV Kishenpur -Samba D/C line inside the substation area near the boundary wall. However, the possibility of creating space for the line reactor exists if the afore mentioned dead-end tower is dismantled and relocated to an open area outside the boundary wall.

Sterlite vide mail 04.03.24 informed that for installation of line reactor at Kishtwar end of 400kV Kishtwar-Samba 400kV line(Quad), no sufficient space is available. However, possibilities may be explored in adjacent additional land where some alternation will be required to accommodate this new Line Reactor. Further as it will be switchable line Reactor, so there will be requirement of 400kV GIS Bay & it will require GIB Bay modification & GIS hall extension. In view of more complexity and additional land requirements at Kishttwar S/s, it is recommended to propose line reactor at Samba end of 400kV Kishtwar-Samba 400kV line(Quad)

Accordingly following reactive compensation is proposed as part of Transmission scheme for Ratle HEP (850MW) in the meeting:

1. 1x63 MVAr Switchable line reactor at Samba end of 400kV Kishtwar-Samba 400kV line(Quad) (165km) [formed after bypassing of 400kV Kishtwar – Kishenpur line (Quad) at Kishenpur and connecting it with one of the circuit of Kishenpur-Samba 400kV D/c line(Quad))
2. 1x80 MVAr Switchable line reactor at each ckt at Jalandhar end of Kishenpur– Jalandhar D/c direct line (Twin) (170km) (formed after bypassing both ckts of 400kV Kishenpur – Samba D/c line (Twin) & 400 kV Samba – Jalandhar D/c line (Twin) at Samba and connecting them together to form Kishenpur– Jalandhar D/c direct line (Twin))
3. 1x63 MVAr Switchable line reactor at Samba end of Samba –Nakodar direct line (Quad) (177km) (formed after bypassing of 400kV Jalandhar – Nakodar line (Quad) at Jalandhar and connecting it with one of the circuit of Samba-Jalandhar 400kV D/c line(Quad Moose), thus forming Samba –Nakodar line (Quad))

In the meeting, Grid-India stated that due to prevailing high voltage issue in hydro pockets, it is recommended that 80MVAr line reactor may be considered in place of 63MVAr line reactor on all above lines. CTU stated that with 80 MVAr line reactor, % reactive compensation will be about 60%, (for S.No .1) & 56% (for S.No 3). CEA & CTU agreed for the same. JKPTCL also agreed for the proposal. POWERGRID also mentioned that bay equipment at Samba vacated after upgradation works at Samba may be considered as regional spare.

Subsequently, the proposal was also deliberated in 49th NR TCC meeting held on 29.03.24, wherein POWERGRID mentioned that on Kishenpur– Jalandhar D/c line with 80MVAr line reactor at Jalandhar end, % reactive compensation would be >70%, which is on higher side, as they have encountered resonance issues for higher order reactive compensation (>70%) in some of the 400kV lines. Therefore, 1x63MVAr line reactor at Jalandhar end on above line should be considered with which reactive compensation would be about 56%.

In the 49th TCC meeting, it was decided that CTUIL and Grid-India may mutually discuss the requirement of reactive compensation on above line and intimate it to NRPC for inclusion in TCC/NRPC MOM. In view of above, CTU requested Grid-India to provide their recommendation for installation of 63 or 80MVAr line reactor at Jalandhar end for Kishenpur– Jalandhar D/c direct line (Twin) (170km).

Grid-India vide mail dated 03.04.24 informed that in view of twin configuration of Kishenpur-Jalandhar D/c line (earlier considered Quad in meeting discussion) and request of POWERGRID, line reactor size may be selected as 63MVAR at 400kV Jalandhar end of 400kV Kishenpur– Jalandhar D/c direct line (Twin) formed with bypassing at Samba by both ckts of 400kV Kishenpur – Samba D/c line (Twin) & 400 kV Samba – Jalandhar D/c line (Twin) at Samba. CTU further solicited views on above modifications (63MVAr in place of 80MVAr line reactor on 400 kV Kishenpur– Jalandhar D/c line) with other stakeholders vide mail dated 03.04.24 for comments on modifications by 04.04.24, however no comments were received. Accordingly, 63MVAr switchable line reactor at Jalandhar end for Kishenpur– Jalandhar D/c direct line (Twin) line is considered.

Accordingly following reactive compensation is agreed as part of Transmission scheme for Ratle HEP (850MW) :

- 1x80 MVAr Switchable line reactor at Samba end of 400kV Kishtwar-Samba 400kV line(Quad) (160km) [formed after bypassing of 400kV Kishtwar – Kishenpur line (Quad) at Kishenpur and connecting it with one of the circuit of Kishenpur-Samba 400kV D/c line(Quad))
- 1x63 MVAr Switchable line reactor at each ckt at Jallandhar end of Kishenpur– Jalandhar D/c direct line (Twin) (170km) (formed after bypassing both ckts of 400kV Kishenpur – Samba D/c line (Twin) & 400 kV Samba – Jalandhar D/c line (Twin) at Samba and connecting them together to form Kishenpur– Jalandhar D/c direct line (Twin))
- 1x80 MVAr Switchable line reactor at Samba end of Samba –Nakodar direct line (Quad) (177km) (formed after bypassing of 400kV Jalandhar – Nakodar line (Quad) at Jalandhar and connecting it with one of the circuit of Samba-Jalandhar 400kV D/c line(Quad Moose), thus forming Samba –Nakodar line (Quad))

Additionally, 400kV bay upgradation works at Samba S/s (for 4 nos. vacated bays due to bypassing) and Kishenpur S/s (for 1 bay) is also agreed in meeting. In view of deliberations held in 26th CMETS-NR meeting as well as present meeting including stakeholder comments, consolidated transmission scheme for evacuation of Ratle HEP is agreed as under :

| Sl. No. | Description of Transmission Element | Scope of work (Type of Substation/Conductor capacity/km/no. of bays etc.) |
|---|---|---|
| 1 | 400 kV Kishenpur-Samba D/c line (Quad)<br><br>(only one circuit is to be terminated at Kishenpur while second circuit would be connected to bypassed circuit of 400kV Kishtwar – Kishenpur line (Quad)) | Length -35 km (Quad) |
| 2 | Bypassing of one ckt of 400kV Kishtwar – Kishenpur 400kV D/c line (Quad) at Kishenpur and connecting it with one of the circuit of Kishenpur-Samba 400kV D/c line(Quad), thus forming **400kV Kishtwar - Samba (Quad) direct line (one ckt)** | |
| 3 | 1x80 MVAr Switchable line reactor at Samba end of 400kV Kishtwar-Samba 400kV line-165km (Quad) [formed after bypassing of 400kV Kishtwar – Kishenpur line (Quad) at Kishenpur and connecting it with one of the circuit of Kishenpur-Samba 400kV D/c line(Quad)) | • 420 kV, 80 MVAr switchable line reactors at Samba S/s end– 1 nos.<br><br>• Switching equipment for 420kV, 80 MVAr switchable line reactors at Samba S/s end – 1 no |

| Sl. No. | Description of Transmission Element | Scope of work (Type of Substation/Conductor capacity/km/no. of bays etc.) |
|---|---|---|
| 4 | Bypassing both ckts of 400kV Kishenpur – Samba D/c line (Twin) & 400 kV Samba – Jalandhar D/c line (Twin) at Samba and connecting them together to form **400kV Kishenpur– Jalandhar D/c direct line (Twin)** | |
| 5 | Bays upgradation works (2000A to 3150A) at Samba end (bays vacated after bypassing of Kishenpur – Samba D/c line (Twin) & 400 kV Samba – Jalandhar D/c line (Twin)) | 400kV Bay upgradation works- 4 nos. bays |
| 6 | 1x63 MVAr Switchable line reactor on each ckt at Jallandhar end of Kishenpur– Jalandhar D/c direct line -170km(Twin) (formed after bypassing both ckts of 400kV Kishenpur – Samba D/c line (Twin) & 400 kV Samba – Jalandhar D/c line (Twin) at Samba and connecting them together to form Kishenpur– Jalandhar D/c direct line (Twin)) | • 420 kV, 63 MVAr switchable line reactors at Jallandhar S/s end– 2 nos.<br>• Switching equipment for 420kV, 63 MVAr switchable line reactors at Jallandhar S/s end – 2 no |
| 7 | 400kV Samba- Jalandhar D/c line(Quad)<br><br>(only one circuit is to be terminated at Jalandhar while second circuit would be connected to bypassed circuit of Jalandhar –Nakodar 400kV D/c line) | Line Length -135 km |
| 8 | 1x80 MVAr Switchable line reactor at Samba end of Samba –Nakodar direct line (Quad) formed after bypassing of 400kV Jalandhar – Nakodar line-177km (Quad) at Jalandhar and connecting it with one of the circuit of Samba-Jalandhar 400kV D/c line(Quad Moose), thus forming Samba –Nakodar line (Quad) | • 420 kV, 80 MVAr switchable line reactors at Samba S/s end– 1 no.<br>• Switching equipment for 420kV, 80 MVAr switchable line reactors at Samba S/s end – 1 no. |
| 9 | Bypassing 400kV Jalandhar – Nakodar line (Quad) at Jalandhar and connecting it with one of the circuit of Samba-Jalandhar 400kV D/c line(Quad Moose), thus forming **400kV Samba –Nakodar line** | |
| 10 | LILO of 400 kV Kishenpur- Dulhasti line (Twin) at Kishtwar S/s along with associated bays at Kishtwar S/s | LILO Length- 10km<br>• 400kV Kishenpur -Kishtwar (LILO section) is on Twin HTLS (with minimum 2100 MVA capacity) configuration |

| | | |
|---|---|---|
| | | • 400kV Dulhasti -Kishtwar (LILO section) is on Twin Zebra configuration<br>• 400kV line bays at Kishtwar – 2 nos. (line bays at Kishtwar S/s end shall be rated accordingly) |
| 11 | Reconductoring of 400 kV Kishenpur-Kishtwar section** with Twin HTLS (minimum 2100 MVA capacity) (formed after LILO of Kishenpur-Dulhasti line at Kishtwar S/s) along with bay upgradation works (2000A to 3150A) at Kishenpur end for above line. | Length – 132km<br>400kV Bay upgradation work- 1 no. bay at Kishenpur end |

***Upto LILO point (LILO of 400 kV Kishenpur- Dulhasti line at Kishtwar S/s) from Kishenpur end***

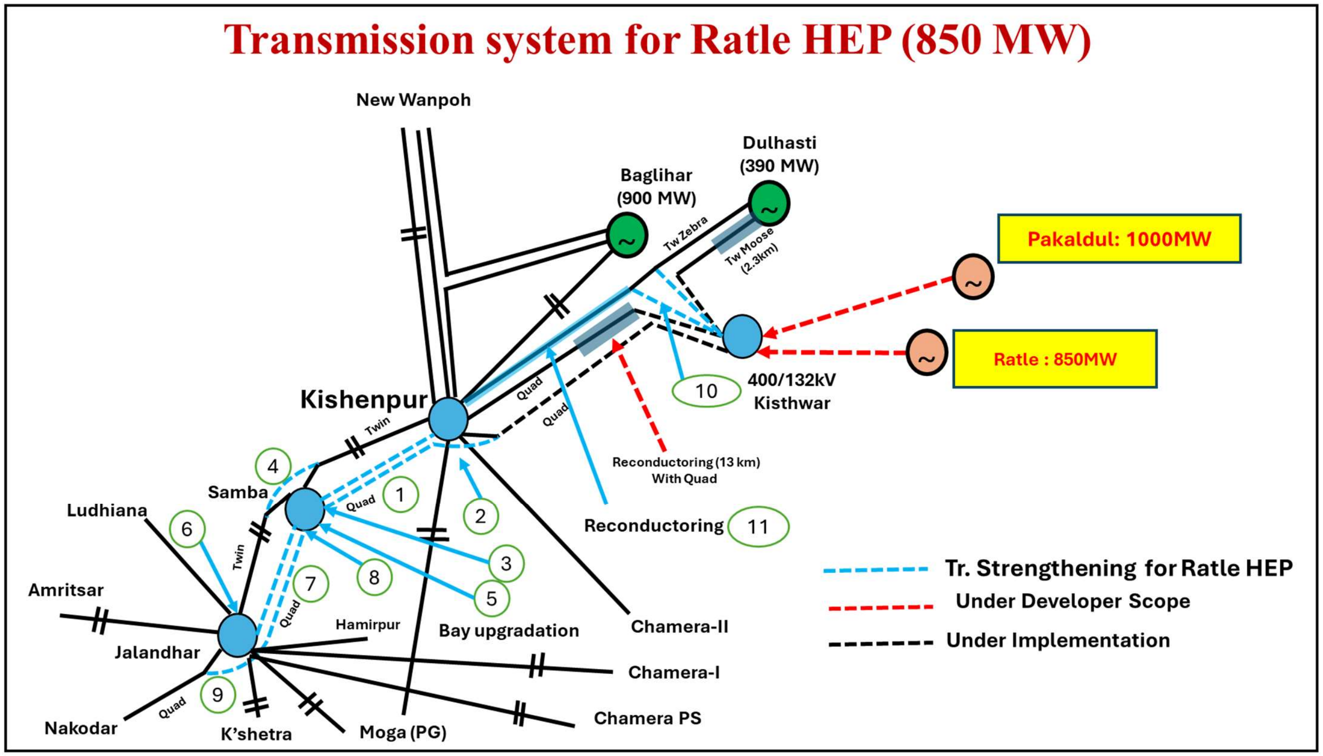


**Fig-3: Transmission scheme for evacuation of power from Ratle HEP**

**Annexure-A**

**List of Participants of 28th Consultation meeting for Evolving Transmission Schemes in NR held on 27.03.2024**

**CEA**

| | |
|---|---|
| Shri Nitin Deswal | Deputy Director |

**SECI**

| | |
|---|---|
| Shri R.K.Agarwal | Consultant |
| Smt. Anita | DGM |

**Grid India**

| | |
|---|---|
| Shri Gaurav Malviya | Manager |

**CTU**

| | |
|---|---|
| Shri V Thiagarajan | Sr. GM (CTU) |
| Shri Kashish Bhambhani | GM (CTUIL) |
| Shri Sandeep Kumawat | DGM (CTU) |
| Smt. Ankita Singh | Ch. Manager (CTU) |
| Shri R Narendra Sathvik | Manager (CTU) |
| Shri Madhusudan Meena | Engineer (CTU) |
| Shri Ganeswara Rao Jada | Deputy Director CEA (on Deputation at CTUIL) |

**PGCIL**

| | |
|---|---|
| Shri Ravindra Nath Gupta | CGM |
| Shri Jagat Ram | GM |
| Shri Mani Shekhar | Sr. DGM |
| Shri Rohit Kumar Jain | DGM |

**RVPNL**

| | |
|---|---|
| Shri Dr. Om Prakash Mahela | Executive Engineer |

**PTCUL**

| | |
|---|---|
| Shri Karanjeet Singh | Assistant XEN |

**JKPTCL**

| | |
|---|---|
| Shri Rajan Gupta | CE |

**UPPTCL**

| | |
|---|---|
| Sh. Satendra Kumar | SE |